

Data Exploration and Preparation

Diploma in DAB – L7

CCT College Dublin

Exploratory Data Analysis

Week 5

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Outline

- Introduction to Missing Data
- Purpose of Data Cleaning
- What does Data Cleaning involve?
- Documenting the Cleaning Process
- Understanding and dropping Missing Data
- Replacing Missing Values with a Constant
- Imputation using Mean, Median or Mode
- Choosing the Appropriate Strategy

Introduction to Missing Data

- Missing data refers to values that are **not recorded, unavailable, or absent** for one or more variables in a dataset.
- **Why it matters in data exploration and preparation?**
    - Missing values can reduce data quality and affect the accuracy of analysis or machine-learning models.
    - They may occur because of data-entry errors, unanswered survey questions, system failures, or unavailable information.
    - During data exploration, missing values should be identified.
    - During data preparation, they can be handled by removing affected rows/ columns, replacing values through imputation, or keeping them when the absence itself is meaningful.
- **Example:**
  If a customer dataset contains ***Age = blank*** for some customers, those blank entries are considered missing data.

Introduction to Data Cleaning

- Data cleaning is an essential stage of data preparation. Before analysing a dataset, we need to check whether the data is complete, valid, consistent and suitable for analysis.
- We focus on four closely related areas, such as
    - **Data Cleaning**
    - **Documenting the Cleaning Process**
    - **Handling Missing Data**
    - **Dropping, Imputation and Flagging**
- The main idea is simple: the quality of the analysis depends heavily on the quality of the data provided to the model.

Purpose of Data Cleaning

- Raw data is rarely ready for immediate analysis. It may contain missing values, duplicate records, inconsistent labels, incorrect data types, obsolete fields or unusual values.
- We emphasise that the database values need both **data cleaning** and **data transformation** before exploration can begin.
- The purpose is to minimise the effect of poor-quality inputs on the final analysis, commonly described as **Garbage In → Garbage Out**.
- **Example:** Imagine a supermarket dataset containing customer ages, product prices and transaction values. If ages contain text values, prices contain negative numbers and customer information is incomplete, any analysis may become misleading.

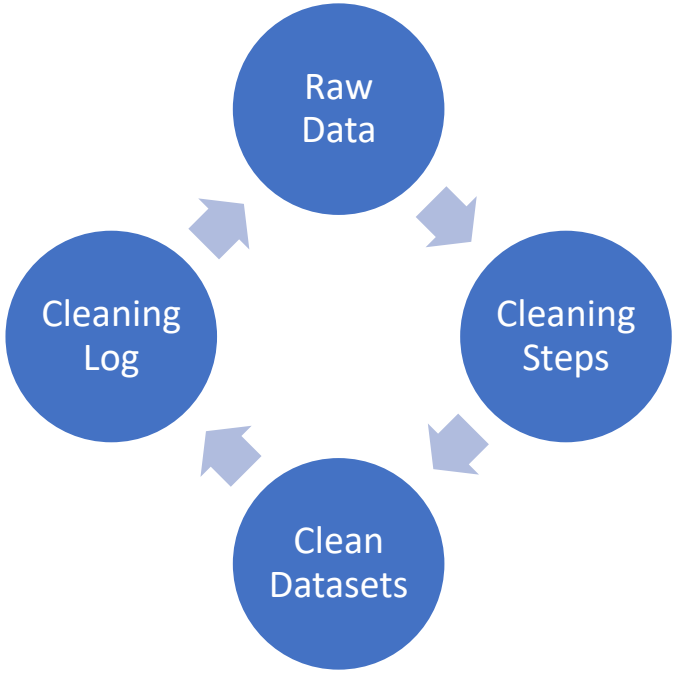
What does Data Cleaning involve?

- Data cleaning means examining the dataset for errors and deciding how those errors should be treated.
- A value may be technically stored in the database but still be incorrect. For example, a postal code such as **06269** can become **6269** when it is incorrectly stored as a numeric field. Similarly, a categorical field such as gender may simply be blank.
- Cleaning involves asking questions, such as
- **Is this value valid?**
- **Is the data type correct?**
- **Does this value make sense?**
- **Is information missing?**

Documenting the Cleaning Process

- Data cleaning should be documented so that another person can understand what was changed and why.
- A good cleaning record should explain
    - **What problem was identified**
    - **Which variable or records were affected**
    - **What action was taken**
- For example: “Three missing age values were replaced using the median age because the distribution was skewed.”
- Documentation improves transparency, reproducibility and accountability.

Recommended workflow



Should we drop missing data?

- One possible approach is to delete rows containing missing values. However, this can be risky.
- If we delete incomplete records, it may remove valuable information from the remaining fields. The pattern of missingness may also be systematic rather than random.
- In the example provided, even a relatively small proportion of missing values could lead to a very large loss of complete records.

```
df = df.dropna()
```

Customer	Age	Income
A	29	€42,000
B	Missing	€51,000
C	44	€63,000

- If ***Customer B*** is deleted, the income information is also lost.
- **Question:** Does deleting the record create more damage than keeping it?

Replacing Missing Values with a Constant

- A simple alternative is to replace missing values with a fixed value selected by the analyst.
- For numeric variables, a constant such as **0** may be used. For categorical variables, a label such as **“Missing”** or **“Unknown”** may be inserted.
- This method replaces missing numeric values with **0.0** and missing category values with **“Missing”**.
- **Caution**
- A constant must not accidentally imply real meaning.
- For example, replacing a missing age with 0 may suggest that the customer is a newborn. A separate “Missing” category may be safer when the distinction matters.

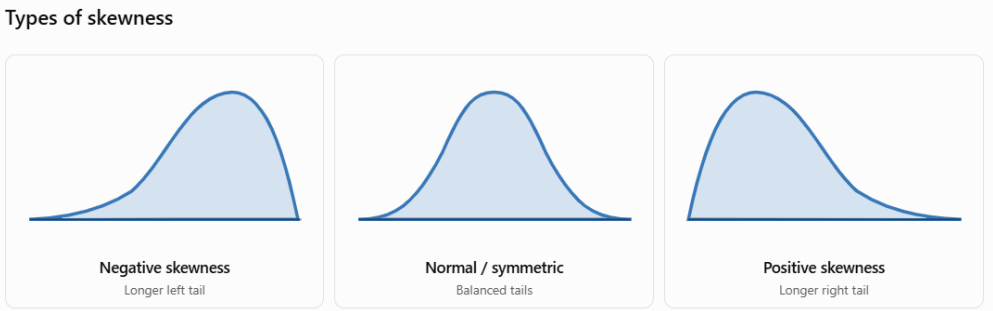
Imputation using Mean, Median or Mode

- Another approach is **imputation**, where missing values are replaced with a reasonable estimate.
- For categorical data, the **mode** may be used.
- For numerical data, the **mean** or **median** may be suitable.
- We replace the missing brand value with mode as **US**, while the missing cubic-inch value is replaced by the mean value **200.65**.
- **Mean = 285.25**, Cubic-inches: 350, missing, 302, 400, 89
- **Median = 326**, After imputation: 89, 302, 350, 400
- Imputation keeps the record in the dataset, but the replacement value is an estimate rather than an original observation.

	mpg	cubicinches	hp	brand
1	14.000	350	165	US
2	31.900		71	Europe
3	17.000	302	140	US
4	15.000	400	150	
5	37.700	89	62	Japan

Limitations

Mean and Mode Imputation



- Mean or mode imputation can be convenient, but it is not always the best solution.
- The mean may not represent a good “typical” value. Replacing many observations with the same value can also reduce variation in the dataset artificially, which may make statistical confidence appear stronger than it really is.
- **For Example:** If salary data is strongly skewed by a small number of very high earners, the mean may be much higher than the salary of a typical employee.
- In that case, the **median** may provide a more appropriate replacement.
- **Main Principle**
- Choose the imputation method according to the variable and its distribution.

Flagging Missing or Imputed Values

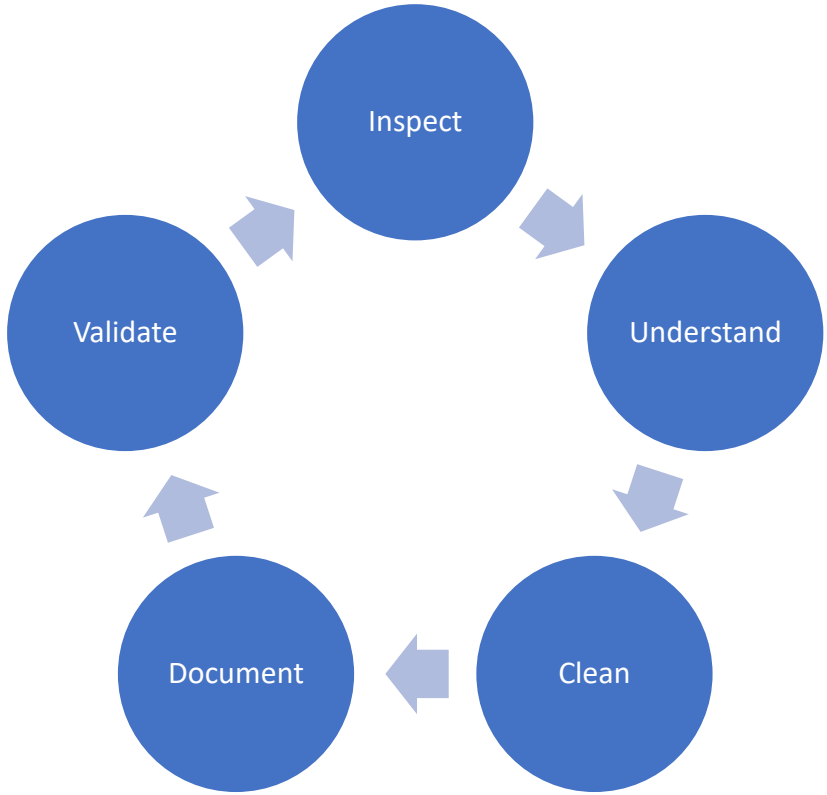
Age	Age_Missing
25	0
31	1
38	0

- A useful technique is to create an additional variable showing whether the original value was missing.
- **For example:** Here, 31 may be an imputed value, while ***Age_Missing = 1*** tells us that the original observation was unavailable.
- Flagging helps to preserve information about the original condition of the dataset. It is particularly useful when missingness itself could be meaningful.
- For example, customers who do not provide income information may behave differently from customers who do.
- **Good Practice**
- Impute the value when appropriate, but keep a record that it was originally missing.

Choosing the Appropriate Strategy

- There is no single correct solution for every missing-data problem.
- **Dropping** may be acceptable when only a very small number of observations are affected and their removal is unlikely to introduce bias.
- **Imputation** may be preferable when retaining the observation is important, but the replacement method should be appropriate for the variable and distribution.
- **Flagging** is useful when the fact that a value was missing may itself contain information.
- Advanced imputation methods estimate missing values using interpolation, regression, classification trees, or regression trees, based on neighbouring values or relationships with other attributes in the dataset.

Final Cleaning Principle



Resources/ References

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- Some images are used from Google search repository (https://www.google.ie/search) to enhance the level of learning.

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